



Direct formation of amphiphilic crosslinked networks based on PVP as a marine anti-biofouling coating

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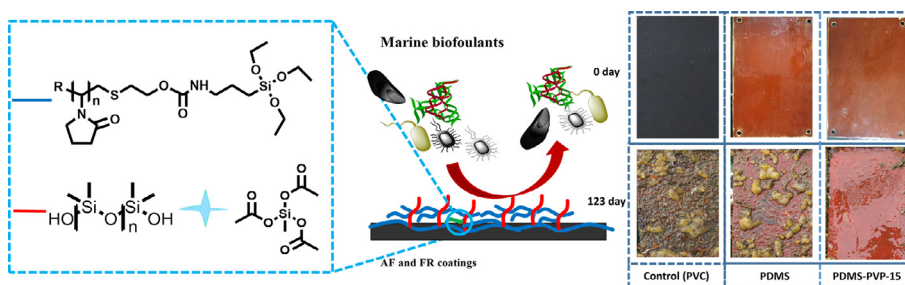
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HIGHLIGHTS

- A novel amphiphilic network coating is based on the physicochemical stable PVP.
- The coating can be directly formed by chemical crosslinking at room temperature.
- The coating shows excellent anti-fouling performance in marine field.

GRAPHICAL ABSTRACT



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ABSTRACT

In this work, we developed a novel amphiphilic coating by incorporating poly (N-vinylpyrrolidone) (PVP), a highly hydrophilic polymer with strong physicochemically stability, into polydimethylsiloxane (PDMS) matrix. PVP was firstly functionalized with triethoxysilane and then directly crosslinked with PDMS to form an amphiphilic network coating at room temperature. Results suggested that the hydrophilicity of PDMS matrix was significantly improved after the modification. Moreover, compared to pristine PDMS, PDMS-PVP coatings could efficiently prevent the adhesion of bacteria and *Navicula* diatoms (both 99% reduction), as well as resist the attachment of mimetic barnacle (82% reduction). More importantly, PDMS-PVP coatings also showed excellent anti-biofouling performance for at least 4 months in the marine field tests. This work provides a novel strategy to prepare high-performance anti-biofouling coatings, which are expected to advance the development of environmentally-friendly coatings in marine applications.

1. Introduction

Marine biofouling can substantially increase the economic and

environmental burdens, thus it has become a significant challenge in the global maritime industry [1–3]. For example, the estimated overall cost associated with hull fouling can achieve up to 260 million per

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annum for an US Navy fleet [4,5]. Marine biofouling is the undesirable attachment and accumulation of various marine organisms on the submerged surfaces [6,7]. These biofoulings can increase hydrodynamic drag, decrease turbine performance, and lead to significantly higher fuel consumption in order to maintain the required speed [4,8–10]. It has been reported that the increased roughness of a heavily fouled ship hull can cause up to 86% of powering penalties at cruising speed, and increase up to 45% of fuel costs [11]. Currently, the primary strategy to resist biofouling is to use biocide-containing paints to kill fouling organisms or inhibit their adhesion [12,13]. However, the regulations of these paints are getting more and more restricted, because their toxicity can simultaneously impact on other non-target species. This is highly unfavorable to environment and is a threat to humans consuming contaminated seafoods [14,15].

Many efforts have been made to develop 'green' alternatives to biocide-based technologies, such as the design of non-toxic marine antifouling (AF) or fouling-release (FR) coatings [16–20]. For example, polydimethylsiloxane (PDMS) is a commonly used FR coating to release biofoulings. Because PDMS materials possess lower surface energy and elastic modulus [21–23], inducing weak adhesion strength of hard foulers, such as barnacle and mussels [24,25]. However, due to their intrinsic hydrophobic property, PDMS materials fail to resist the attachment of marine slime layers, which are mainly composed of bacteria and diatoms [26,27].

Currently, hydrophilic polymers, such as polyethylene glycol (PEG), are widely used to physically blend or chemically modify PDMS to form amphiphilic anti-biofouling coatings [16,28–35]. These amphiphilic coatings possess bifunctional nature to incorporate hydrophobic and hydrophilic segments [5,36], and present 'ambiguous' surfaces to the complex protein or glycoprotein adhesives against their attachment [24,37,38]. However, based on currently used hydrophilic polymers, the amphiphilic coatings using traditional preparation methods suffer from several problems. First, the instability of commonly used hydrophilic polymers may influence the long-term applications of the coatings. For example, PEG is susceptible to oxidative damage, because the cleavage of ether bonds in PEG polymers can be readily induced by oxidation [39,40]. It has been reported that after several months in biological environments, the chemical structure change of PEG coatings could significantly reduce their anti-biofouling capability [41,42]. Second, amphiphilic polymers are commonly prepared by the complicated chemical steps in harsh conditions. For example, PEG-modified PDMS materials are usually prepared by surface-initiated atom transfer radical polymerization (ATRP) or reversible addition-fragmentation chain transfer polymerization (RAFT) methods, both requiring complicated chemical synthetic steps for preparation of block polymers and chain transfer agents under oxygen-free condition [43–45]. Third, the prepared amphiphilic polymers are commonly unable to be directly crosslinked as a coating layer, so they need to be physically blended in the substrate matrix (such as PDMS) in order to form paint coatings at ambient conditions. So it is conceivable that these amphiphilic coatings exhibit a poor long-term stability.

Recently, poly(*N*-vinylpyrrolidone) (PVP) has become an attractive material due to its high hydrophilicity and biocompatibility. When grafted on glass or silicon surfaces, PVP show a strong ability to resist protein and cell adhesion, because its hydrophilic pyrrolidone units can efficiently eliminate the electrostatic and hydrophobic interactions with biomacromolecules [46–51]. Moreover, PVP possesses excellent chemical stability and is difficult to be oxidized. It has been reported that PVP materials could maintain their structure against oxidation even in 3% of H_2O_2 solution for 100 h at 60 °C [52,53].

In this work, we presented a novel method to develop a stable coating for marine anti-biofouling applications. This method could directly form an amphiphilic crosslinked polymer network coating, which was based on PVP with strong physicochemical stability (Fig. 1). PVP was functionalized with triethoxysilane (PVP-IPTS) to directly crosslink with PDMS matrix at room temperature. The amphiphilic PDMS-PVP

coating exhibited an excellent performance of biofouling resistance in both laboratory and marine field, indicating its potential to serve as an environmentally-friendly marine anti-biofouling coating.

2. Materials and methods

2.1. Materials

Dihydroxyl-terminated polydimethylsiloxane (HO-PDMS-OH) ($M_w = 4200 \text{ g mol}^{-1}$) was purchased from Alfa Aesar, USA. (3-Isocyanatopropyl)-triethoxysilane (95%) and methyltriethoxysilane (METES) (96%), isopropyl alcohol (IPA), 2,2'-Azobis(2-methyl-N-(2-hydroxyethyl)propionamide (AMPAHE), tetrahydrofuran (THF), dichloromethane (CH_2Cl_2), diethyl ether, acetone were all obtained from Heowns, China. Dibutyltin dilaurate (DBTDL, 97.5%) was purchased from j&k, China. *N*-vinylpyrrolidone (NVP, 99% stabilized with NaOH) was obtained from Adamas and purified to remove the inhibitors by distillation under reduced pressure before use. 2-mercaptoethanol (MCE, 98%) was obtained from TCI, Japan. Human plasma fibrinogen was purchased from Sigma-Aldrich (Milwaukee, WI). Fluorescent isothiocyanate labeled bovine serum albumin (FITC-BSA) and tris (hydroxymethyl) aminomethane (Tris) crystallized free base (molecular biology grade) were both purchased from Solarbio. *Navicula* diatoms were obtained from Science and Technology on Marine Corrosion and Protection Laboratory, Qing dao, China. Deionized water was used in all experiments.

2.2. Synthesis of PVP with a hydroxyl-bearing chain end (PVP-OH)

As shown in Fig. 1b, PVP-OH was synthesized by free radical polymerization of NVP according to previous literature report [54]. Briefly, NVP (15 g, 135 mmol), AMPAHE (0.39 g, 1.35 mmol), and MCE (0.53 g, 6.75 mmol) were dissolved in 200 mL IPA and degassed with N_2 for 10 min. The reactants were polymerized for 24 h at 85 °C, and then IPA was removed under reduced pressure. The product was precipitated in 200 mL diethyl ether. The solid was collected and then dissolved in 15 mL of dichloromethane and precipitated again in 200 mL diethyl ether. Lastly, the product was transferred into a round-bottomed flask and dried in a vacuum oven at 80 °C overnight. The ^1H NMR, FTIR and GPC results were shown in Figs. S1 and S3.

2.3. Synthesis of the triethoxysilane-functionalized PVP (PVP-IPTS)

(3-Isocyanatopropyl)-triethoxysilane (0.8 g) and DBTDL (0.0063 g) were slowly added to a mixture solution of PVP-OH (8.0 g) and THF (10 mL). Then, the solution was stirred at 65 °C for 3 h to obtain PVP-IPTS polymers. This product was used directly in the next step without any pretreatment. The ^1H NMR and FTIR results were shown in Figs. S2 and S3.

2.4. Preparation of amphiphilic crosslinked network coatings

Glass slides were ultrasonically cleaned in acetone and ethanol for 15 min respectively and rinsed with deionized water. Meanwhile, a solution composed of HO-PDMS-OH (5.0 g), METES (0.1 g) and DBTDL (50 mg) was dissolved in THF (25 mL) and stirred constantly for 0.5 h at room temperature for connection layers. Similarly, PVP-IPTS at different concentrations (5 wt%, 10 wt% and 15 wt%) were added to the solution above, and the solutions were stirred constantly for 0.5 h at room temperature for prepolymers (Table 1). Subsequently, the PDMS-PVP-X coatings were prepared by a three-step procedure as follows. (1) The solution for connection layer was painted onto the glass slides. Then, the coatings were gradually formed and dried at room temperature for 24 h, and they were annealed at 120 °C for 12 h for the accelerative removal of the volatile solvents. (2) The operation of (1) was repeated to form the second layer. (3) Lastly, the solution containing

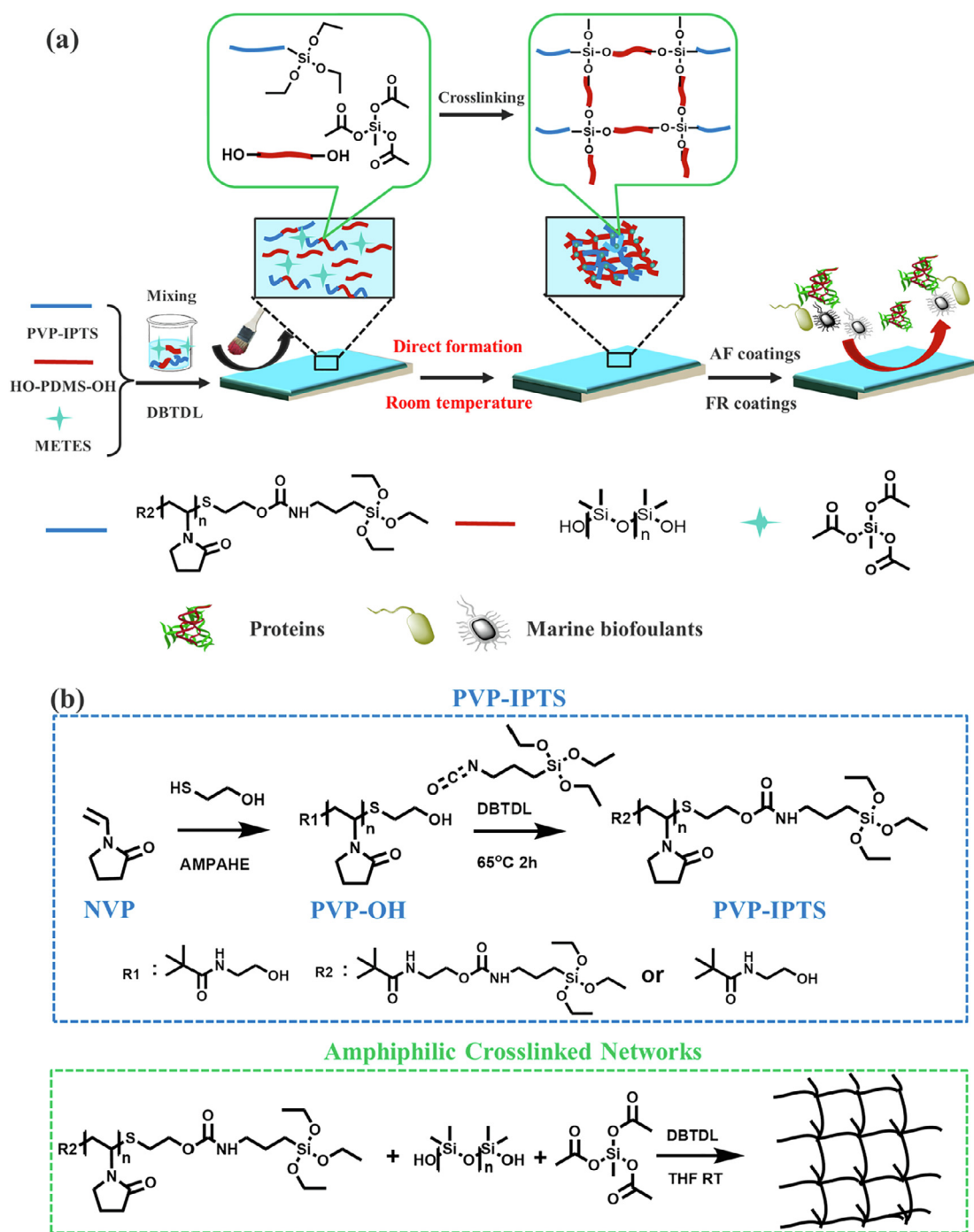


Fig. 1. (a) Schematic illustration for the design of the amphiphilic network coating on glass. (b) The synthesis route of the triethoxysilane-functionalized PVP and the amphiphilic crosslinked networks.

Table 1

Formulations of the crosslinked polymer network coatings.

	PDMS(wt %)	PVP-IPTS (wt %)	Solvent and crosslinking agent (wt%)
PDMS-PVP-5	50	2.5	47.5
PDMS-PVP-10	50	5	45
PDMS-PVP-15	50	7.5	42.5
PDMS	50	0	50

prepolymers was painted on the connection layer and crosslinked and dried at room temperature for 12 h. The final amphiphilic crosslinked network PDMS-PVP-X coatings were obtained at 120 °C after 12 h to remove the solvents. PDMS-PVP-X coatings were represented by PDMS-PVP-5, PDMS-PVP-10 and PDMS-PVP-15, respectively. The detailed compositions of these coatings were listed in Table 1.

2.5. Characterizations

¹H-NMR spectra were conducted using a Bruker Avance 500 MHz (25 °C). The polymers were dissolved in deuterated chloroform (CDCl₃) or water (D₂O), and all chemical shifts were recorded in ppm (δ). The

surface morphologies of the samples were examined by SEM analysis (SIRISON, FEI Co., Ltd). FTIR spectra were obtained using a Thermo NICOLET 6700 with a scan range of 400–4000 cm^{-1} . Gel permeation chromatography (GPC) tests were performed with a Shimadzu LC-20AD, GPC system. Polystyrene was used as the standard, and THF was used as an eluent at a flow rate of 1.0 mL/min. The surface hydrophilicity and hydrophobicity were investigated by the water contact angle measurements. The contact angles (θ s) (water and methylene iodide (CH_2I_2)) were measured with an optical θ meter (Shanghai Zhongchen Digital Technology Apparatus Co. JC2000D1) at room temperature. The volume of the tested water droplet was 5.0 μL . Five different locations in each sample were chosen to measure the contact angles, and the average value was taken as θ . The surface energies were calculated according to the Owens-Wendt method [55]. The measurement was performed at room temperature and 50% relative humidity. The surface free energy of the coating was calculated by the measurement of θ with two different probe liquids (water and CH_2I_2) with known surface tensions [56], according to the Owens-Wendt method [52] [Eqs. (1) and (2)]:

$$\gamma_{LV}(1 + \cos\theta) = 2(\gamma_{SV}^d \gamma_{LV}^d)^{1/2} + 2(\gamma_{SV}^p \gamma_{LV}^p)^{1/2} \quad (1)$$

$$\gamma_{SV} = \gamma_{SV}^p + \gamma_{SV}^d \quad (2)$$

The superscripts of d and p were the dispersion and polar components of surface free energy, respectively; γ_{LV} was the interfacial tension at liquid-vapor and γ_{SV} was the interfacial tension at solid-vapor interfaces; and θ was the static contact angle at solid-liquid-vapor interfaces.

Advancing (θ_{adv})/Receding contact angles (θ_{rec}) were determined using the inflation/deflation method. The contact angle hysteresis ($\Delta\theta$) was calculated according to the formula: $\Delta\theta = \theta_{\text{adv}} - \theta_{\text{rec}}$

2.6. Anti-sludge adhesion tests

Model sludge (Dyed resin powders, 0.5 g) was sprayed on the pristine PDMS and PDMS-PVP-10 coatings. After 2 h, they were rinsed by continuous water flow. Next, natural sludge obtained from the lake in Tianjin University was also tested. The samples were exposed in an aqueous suspension of sludge (30 g in 50 mL water) for 4 h, and placed on a shaker for 3 min (shaking rate 80 r min^{-1}) to remove the unattached sludge.

2.7. FITC-BSA adsorption tests

2 mg/mL of FITC-BSA solution was prepared in PBS. PDMS-PVP-X and control PDMS coatings (0.7 cm in diameter) were incubated in FITC-BSA solution in dark for 2 h. Then, these coatings were rinsed by deionized water thoroughly, and observed using an inverted microscope. Protein adsorptions on coating surfaces were measured using a fluorospectrophotometer.

2.8. ELISA tests

An enzyme-linked immunosorbent assay (ELISA) was used to measure fibrinogen (Fg) adsorption on the PDMS-PVP-X materials [57]. Briefly, samples (0.7 cm diameter) were placed into individual well of a 24-well tissue culture plate. Then 0.5 mL of fibrinogen solutions (1 mg/mL) from human plasma were added into each well and the culture plate was incubated at 37 °C for 90 min. After rinsed, the samples were incubated with BSA (1 mg/mL in PBS) for 90 min at 37 °C. Then they were transferred in 500 μL of horseradish peroxidase (HRP)-conjugated anti-fibrinogen solution (5.5 $\mu\text{g/mL}$) and incubated for 30 min at 37 °C. Subsequently, 500 μL of citrate-phosphate buffer (0.1 M, pH = 5.0) containing 1 mg/mL chromogen o-phenylenediamine (OPD) and 0.03% hydrogen peroxide (H_2O_2) was added in each well for 20 min at 37 °C in the dark. 500 μL of H_2SO_4 (1 M) was introduced in each well to stop the

enzyme-induced color reaction, and the absorbances of supernatant in each sample were determined at 490 nm by a Tecan Infinite M200 Pro multimode reader. The relative adsorption value = optical density (OD) of the sample/OD of control PDMS material.

2.9. Bacterial adhesion tests

The sterilized PDMS-PVP samples (0.7 cm diameter) were put in a 24-well cell-culture plate, and they were added by 100 μL bacterial suspension (10^6 CFU mL^{-1}). After incubation at 37 °C for 4 h, their surfaces were gently rinsed three times with 1 mL of PBS. 100 μL of the last washing fluid using an ultrasonic cleaner was dispersed in the beef-protein medium, and cultured for 24 h at 37 °C [58].

2.10. Mimetic measurement of barnacle adhesion strength

In order to evaluate the adhesion of marine biofouling on novel coatings, the mimetic measurement of barnacle adhesion strength was used. According to American Society for Testing and Material (ASTM) D 5618, the PosiTest produced by DeFelsko was used to simulate the barnacle. The stud was stuck on the coatings by epoxy. Force at a rate of approximately 4.5 N s^{-1} was applied to a stud edge plate at the base parallel until the stud was detached from the surfaces. Adhesion strength in shear was calculated by dividing the measured force, which was required to remove the stud by the basal area. Error bars represented the standard error of the mean.

2.11. Attachment of *Navicula parva* diatom

Three slides were equilibrated in filtered artificial seawater (ASW) for 72 h prior to the assays. *Navicula parva* diatom suspension was prepared in the filtered and sterilized seawater with a final concentration of 1×10^6 mL^{-1} . 10 mL of cell suspension was added in Quadriperm dishes to test PDMS-PVP and pristine coating surfaces for 2 h or 24 h. Then, the samples were rinsed by ASW to remove the unattached cells, and then immediately subjected to an optical microscope observation (Nikon Eclipse Ti-s). Numbers of attachment diatoms were counted at 30 fields of view on each replicate slide to provide the statistical average number with 95% confidence interval.

2.12. Marine field tests

Carbon steel plates (350 mm $\times$ 100 mm $\times$ 1.5 mm) were coated with samples at 25 °C and 53% of relative humidity. The thickness of the coatings ranging from 250 to 300 μm was controlled by the weight of the polymer solutions. The panels were exposed to seawater at a depth of 0.2–2.0 m from a stationary experimental raft from May 2018 to September 2018 at the inner Qingdao bay (120°30'E, 35°35'N) in China, where fouling organisms are diversified, and biofouling pressure is heavy. The plates were taken out of the sea periodically and washed carefully with seawater to observe the deposition of sea organisms and seaweed on the plates. A polyvinyl chloride (PVC) panel without any coating was used as the control sample. The anti-biofouling activity was evaluated according to American Society for Testing Materials (ASTM D6990-05 (2011)).

3. Results and discussion

3.1. Direct formation of amphiphilic crosslinked network coatings

PVP has been reported that could efficiently inhibit the attachment of proteins and bacteria thanks to its strong hydrophilicity. In this work, due to the lack of functional groups, we introduced a functional group to the terminals of PVP polymer using triethoxysilane in the polymerization process, as shown in Fig. 1b. Firstly, using AMPAHE as the initiator, PVP-OH polymers were obtained by free radical

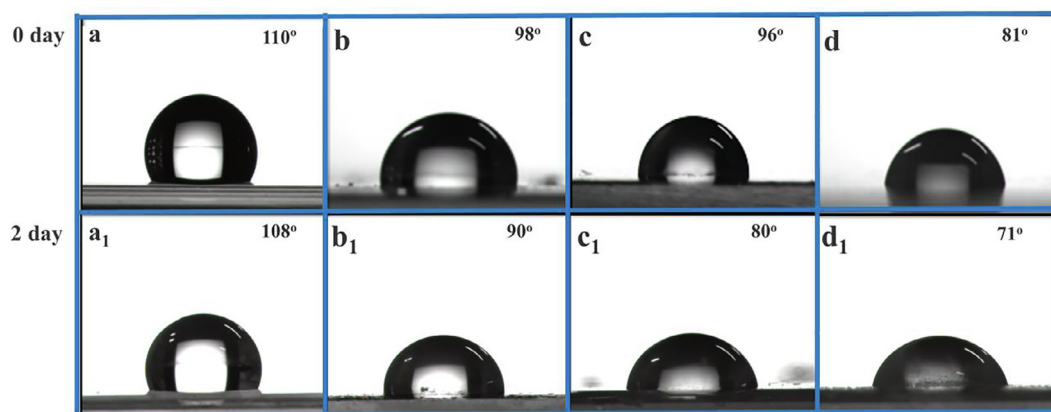


Fig. 2. Water contact angles of (a) PDMS, (b) PDMS-PVP-5, (c) PDMS-PVP-10 and (d) PDMS-PVP-15 coatings before and after 2-day exposure to water.

copolymerization of NVP and 2-mercaptoethanol. The structure of the product was confirmed by ^1H NMR, FTIR and GPC (Figs. S1 and S3). As shown in Fig. S1A–B, the peaks at 2.7 and 2.75 ppm ('c' peak) corresponded to the methylene group of 2-hydroxy-1-ethanethiol (MCE), demonstrating the successful reaction of NVP and MCE for the intermediate product PVP-OH. The molecular weight (M_n) of the PVP-OH was 1666 Da and the PDI was 1.03. Secondly, PVP-OH polymers were reacted with an isocyanate silane coupling agent, obtaining the triethoxysilane functional group in the terminal of PVP (PVP-IPTS). In Figs. S2 and S3, the proton peak at δ 0.5–0.9 corresponded to ethoxy in the end of PVP-IPTS and the new absorption peaks at 1088 cm^{-1} assigned to Si–O–C stretching, demonstrating the successful connection of triethoxysilane functional group to PVP-OH. Finally, PVP-IPTS could directly crosslink with HO-PDMS-OH at room temperature in the presence of crosslinking agents. Three different contents of PVP-IPTS (5%, 10%, and 15%) were prepared in this work.

3.2. Characterization of coating

To examine the wettability of these amphiphilic network (PDMS-PVP) coatings, their water contact angles were evaluated and shown in Fig. 2. In Fig. 2a, the water contact angle of pristine PDMS coating was $\sim 108^\circ$, indicating its high hydrophobicity which is consistent with previous reports. In contrast, the hydrophilicity of PDMS-PVP coatings could be significantly improved and was in direct proportion to the contents of PVP-IPTS (the contact angle decreased from 99° to 84° with 5% to 15% of PVP-IPTS), as shown in Fig. 2b–d. When the content of PVP-IPTS increased to 20%, the mechanical strength of amphiphilic coatings decreased, leading to their weak adhesion with the substrates. After 2-day exposure to water, there was no obvious change in the contact angles of PDMS surface, while those of PDMS-PVP surfaces were all significantly decreased, indicating the increased hydrophilicity of these surfaces. In addition, the advancing/receding contact angle results showed the reduction of the hysteresis contact angles after introducing PVP segments (Table S1 and Fig. S4). Moreover, the surface free energies of these coatings were calculated and studied. It has been reported that barnacle attachment could be efficiently inhibited by the coatings with lower surface free energy ($< 30\text{ mN/m}$) [59]. As shown in Table 2, the addition of PVP-IPTS did not significantly increase the

coating surface free energy (from 25.78 to 30.35 mN/m), suggesting that PDMS-PVP coatings could maintain FR performance.

We further observed the surface morphology of these coatings using SEM, as shown in Fig. 3. Compared with pristine PDMS and PDMS-PVP-5 coatings (Fig. 3a and 3b), the uniformly distributed protuberances and cavities with the diameter of $2\text{--}4\text{ }\mu\text{m}$ presented on the surface of PDMS-PVP-10 and PDMS-PVP-15 coatings (Fig. 3c and d). During the crosslinking reaction, the hydrophilic segments (PVP) were assembled to the microspheres in hydrophobic PDMS matrix, resulting phase separation to form these cavity-type vesicles.

3.3. Anti-sludge adhesion tests

When exposed to the marine environment, the submerged surface of ship hull would inevitably contact and get attached by the sludge in the seawater. Therefore, in order to evaluate their ability to resist sludge adhesion, these coatings were firstly tested using non-biological fouling resin powders (Fig. 4). The dyed resin powders (pink) were sprinkled onto the coatings, and then rinsed with continuous water flow to mimic the ocean currents. In contrast to pristine PDMS coatings, the dyed resin powders on PDMS-PVP coating surface could be easily washed away (Fig. 4b). Moreover, after 4 h exposure to natural sludge suspension, PDMS-PVP also showed excellent anti-sludge adhesion properties (Fig. S5).

3.4. Anti-protein adhesion tests

The initial step of biofilm formation is the biomacromolecules adsorption on submerged surfaces [8]. So in this work, these coatings were exposed to two types of representative proteins (FITC-BSA and Fg) to evaluate their ability to resist protein adsorption. As shown in Fig. 5a, there were a large amount of fluorescent proteins attached on the PDMS coating surface (Fig. 5a1). In comparison, PDMS-PVP coatings with increased hydrophilicity showed no observable fluorescence, indicating their ability to resist protein adhesion. In addition, we quantitatively evaluated their anti-protein adhesion performance by ELISA tests. As presented in Fig. 5b, 74% of Fg adsorption on the PDMS-PVP coatings were reduced compared to pristine PDMS coating. It further demonstrated that the novel amphiphilic coatings showed a

Table 2
Contact angle and surface energy of the coating surfaces.

	Water contact angle ($^\circ$)	CH_2I_2 contact angle ($^\circ$)	$\gamma_{\text{sv}}^{\text{d}}$ (mN/m)	$\gamma_{\text{sv}}^{\text{p}}$ (mN/m)	γ_{sv} (mN/m)
PDMS	108 ± 2	70 ± 2	22.74 ± 1.14	0.13 ± 0.05	22.87 ± 1.15
PDMS-PVP-5	99 ± 1	65 ± 1	24.66 ± 0.73	1.12 ± 0.27	25.78 ± 0.51
PDMS-PVP-10	94 ± 3	62 ± 2	25.91 ± 0.96	2.12 ± 0.74	28.03 ± 0.90
PDMS-PVP-15	84 ± 3	61 ± 2	24.56 ± 1.64	5.79 ± 1.63	30.35 ± 0.58

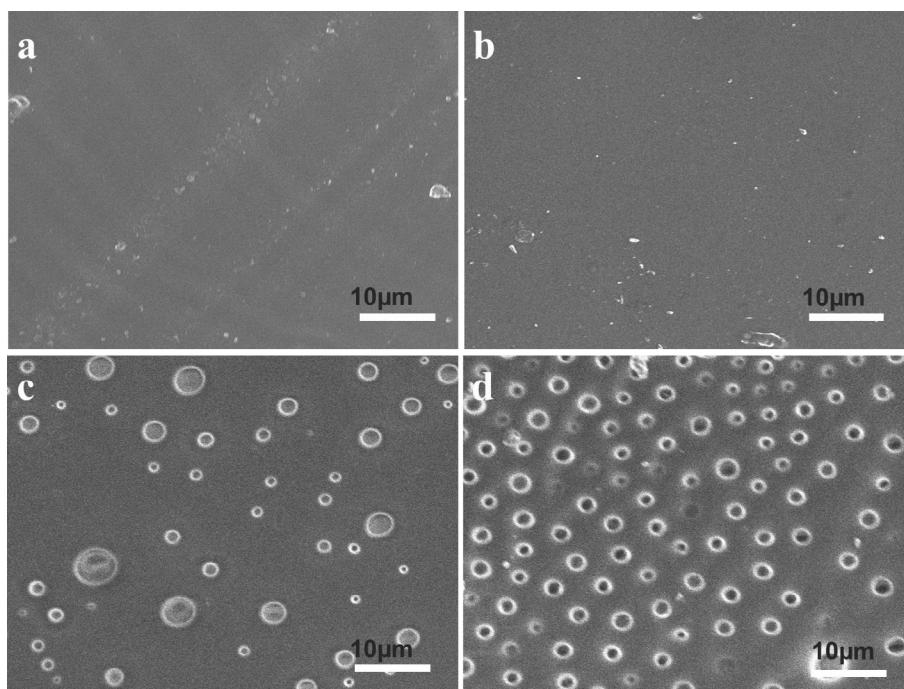


Fig. 3. Surface topography of (a) PDMS, (b) PDMS-PVP-5, (c) PDMS-PVP-10 and (d) PDMS-PVP-15 coatings.

high level of resistance for protein adsorption. This is possibly because that when the coatings were immersed in seawater, the hydrophilic segments could migrate to their surface and form a water film as the barrier.

3.5. Antibacterial adhesion tests

Previous reports have shown that bacterial adhesion is a critical step in biofilm formation. Afterwards, various microorganism attachments follow the bacterial colonies and lead to a complex microbial community [60]. Therefore, we used two representative bacteria, *E. coli* and *S. aureus*, to test antibacterial adhesion of these amphiphilic coatings. As shown in Fig. 6a and b, after 2-h exposure in bacterial suspension there were high-density *E. coli* bacterial colonies on the PDMS coating (47796 cfu/cm^2). In comparison, only few *E. coli* bacterial colonies could be found on the PDMS-PVP-5 (892 cfu/cm^2), PDMS-PVP-10 (840 cfu/cm^2), and PDMS-PVP-15 (446 cfu/cm^2) coatings, indicating ~99% reduction of adhesion. These amphiphilic coatings all exhibited a significant inhibition on bacterial adhesion, and the efficiency showed no obvious colligative properties of PVP contents. The resistance of *S. aureus* was confirmed by Fig. 6. In addition, after the 24-h exposure, PDMS-PVP coatings also performed well to resist the adhesion of

fluorescent *E. coli* (Fig. 7). These results indicated the outstanding antibacterial performance of novel amphiphilic coatings.

3.6. Mimetic measurement of barnacle adhesion strength

Barnacle possesses hard shells and is one of the most harmful bio-fouling to the ship hulls. The pseudobarnacle adhesion is a widely used mimetic measurement to evaluate the barnacle adhesion strength, which can be operated in the lab. According to ASTM D 5618, the stud was used to simulate the hard-shell barnacle, and the epoxy gum was used to simulate the gums secreted by barnacles. As shown in Fig. 8, the tensile forces of PDMS-PVP coating surfaces were significantly decreased compare to that of control PDMS. Meanwhile, the adhesion strength of PDMS-PVP coatings was decreased from 0.11 MPa to 0.02 MPa with 82% reduction. It could be expected that in marine environment the attached barnacles should be easily removed by the hydrodynamic forces at a low ship speed.

3.7. Settlement cells of diatom *Navicula parva*

Diatom is a microscopic unicellular marine colonial alga, which has cell walls impregnated with silica. PDMS materials fail to prevent

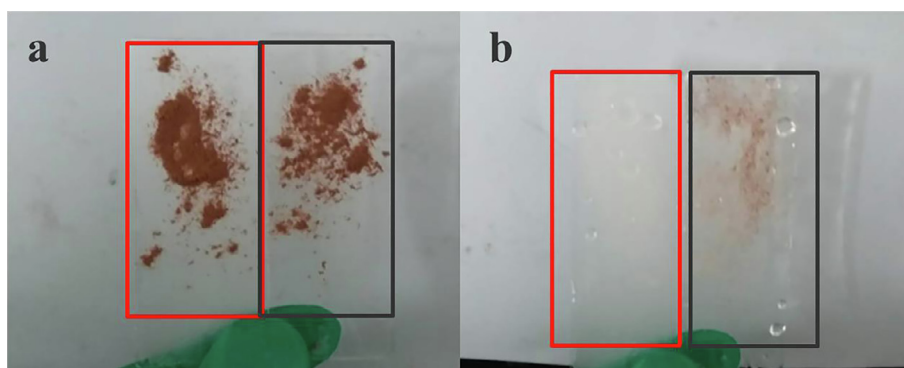


Fig. 4. PDMS-PVP-10 (left) and PDMS (right) coatings (a) before and (b) after water rinsing process.

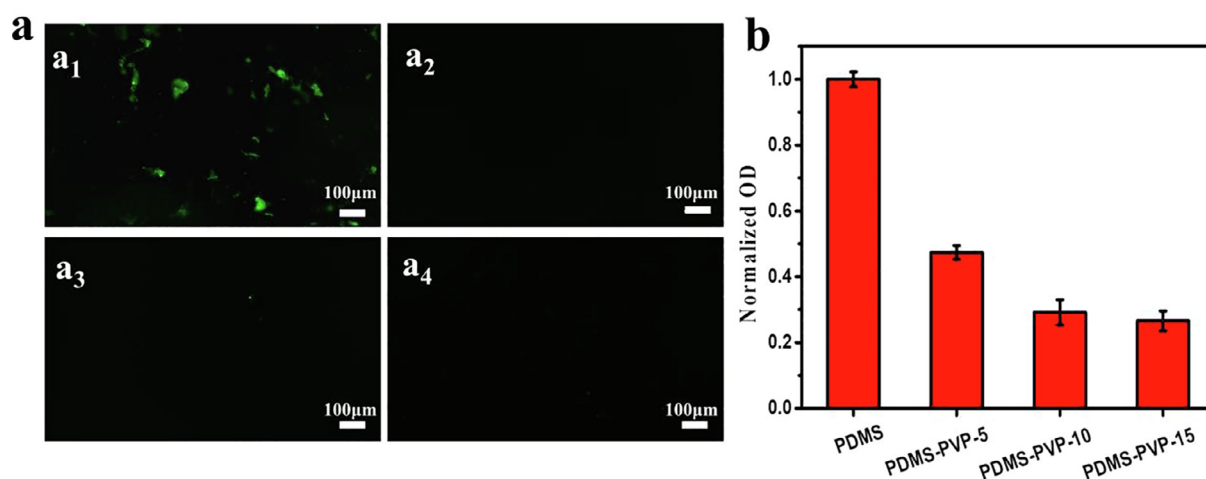


Fig. 5. a) Fluorescent images of BSA-FITC adsorption on (a₁) PDMS, (a₂) PDMS-PVP-5, (a₃) PDMS-PVP-10 and (a₄) PDMS-PVP-15 coatings. Scale bar: 100 μm . b) BSA adsorption by ELISA measurement on these coatings.

diatom adhesion, because PDMS is mainly composed of 'Si-O' and 'Si-C'. Diatoms can easily attach on PDMS surfaces due to the hydrophobic interaction between Silica and 'Si-C' or 'Si-O'. Therefore, the settlement cell numbers of *Navicula parva* diatom were evaluated in this work, as presented in Fig. 9 and Fig. S6. After 2h settlement, the mean

density of diatoms attached on PDMS-PVP-5 coating surface ($784/\text{mm}^2$) was just $\sim 1/10$ as that on the pristine PDMS ($7421/\text{mm}^2$), while almost no diatom attachment on PDMS-PVP-10 and PDMS-PVP-15 coating surfaces could be observed. In addition, as shown in Fig. S6, compared to the pristine PDMS ($10951/\text{mm}^2$), PDMS-PVP coatings exhibited

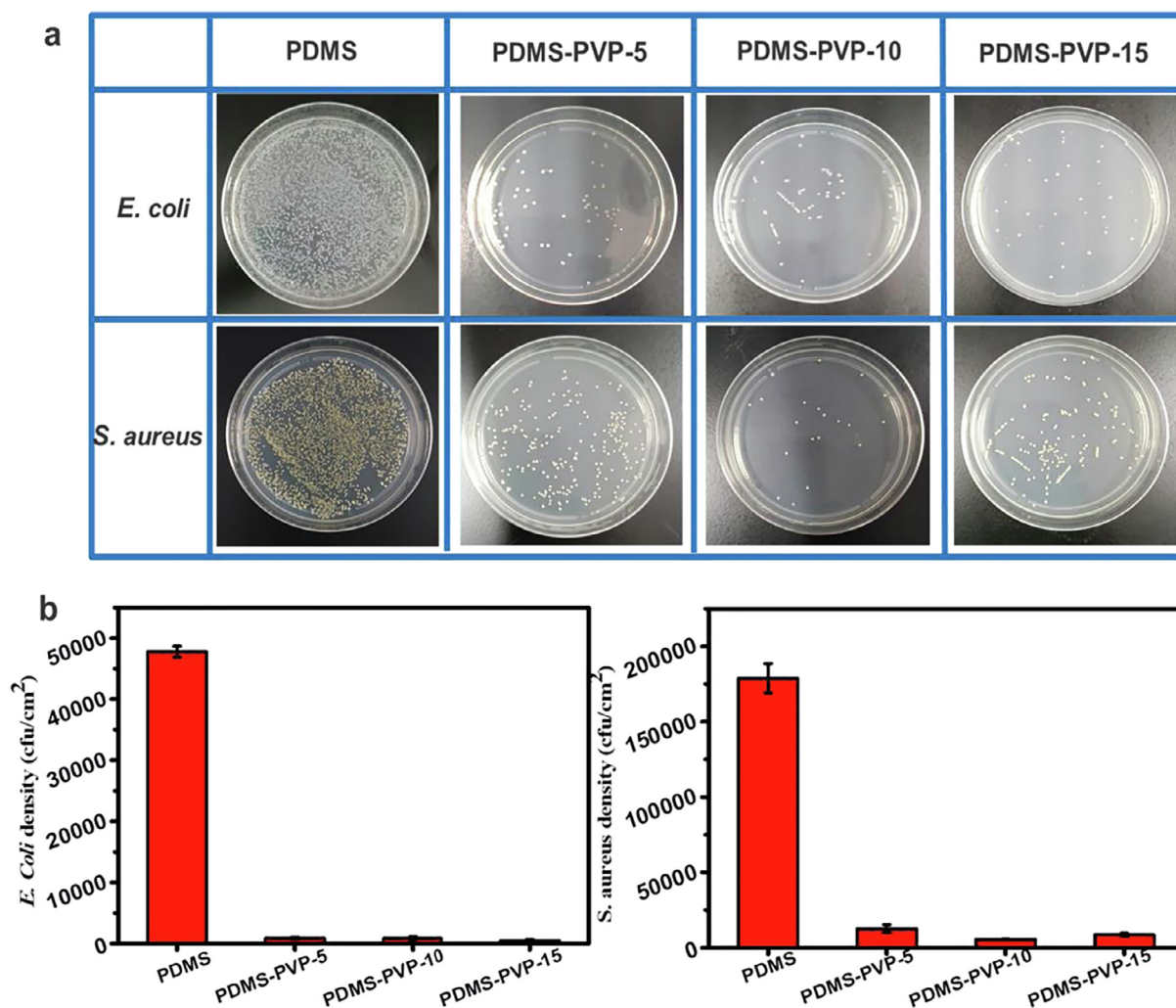


Fig. 6. Antibacterial adhesion tests. (a) Colonial morphology images and (b) numbers of *E. coli* and *S. aureus* attached on the PDMS, PDMS-PVP-5, PDMS-PVP-10, PDMS-PVP-15 coating surfaces.

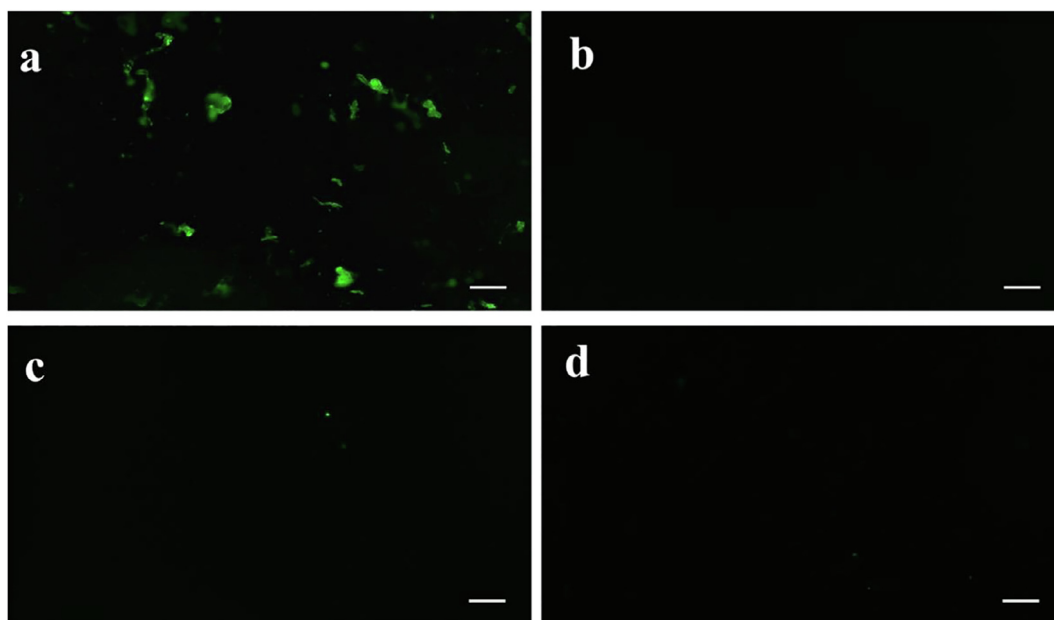


Fig. 7. Fluorescence images of *E. coli* adhesion on (a) PDMS, (b) PDMS-PVP-5, (c) PDMS-PVP-10, (d) PDMS-PVP-15 coating surfaces after 24 h exposure.

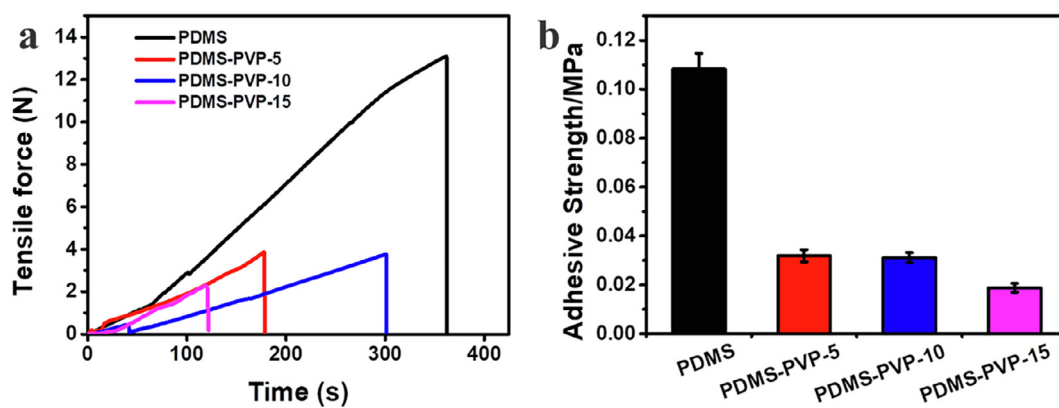


Fig. 8. a) The tensile forces of pseudobarnacle adhesion test during the different periods and b) adhesive strength in the pseudobarnacle adhesion test.

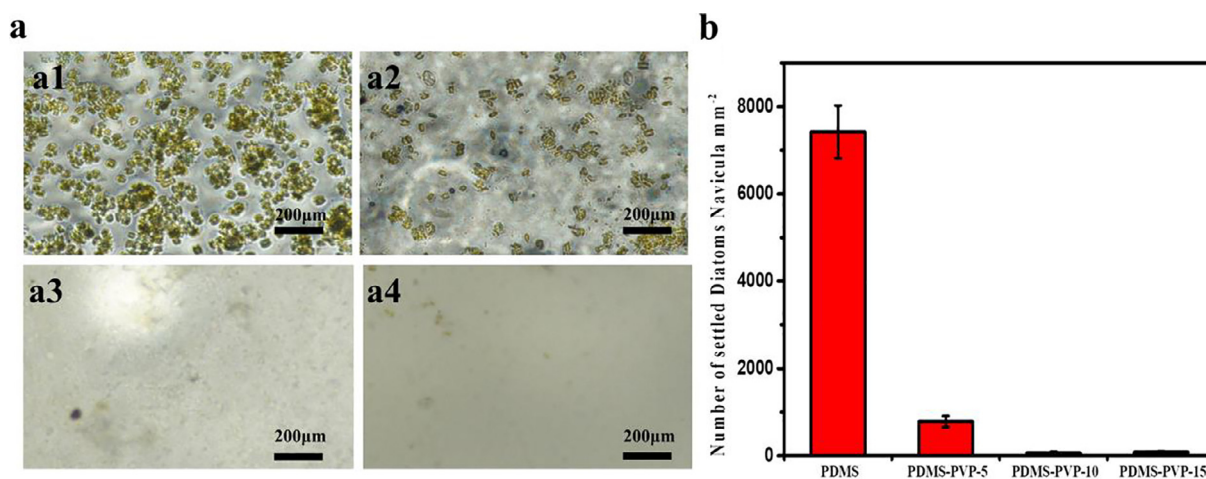


Fig. 9. *Navicula* diatom adhesion. (a) Microscope images of (a₁) PDMS, (a₂) PDMS-PVP-5, (a₃) PDMS-PVP-10, (a₄) PDMS-PVP-15. (b) Mean numbers of diatoms (mm^{-2}) attached to these coatings after 2 h settlement.

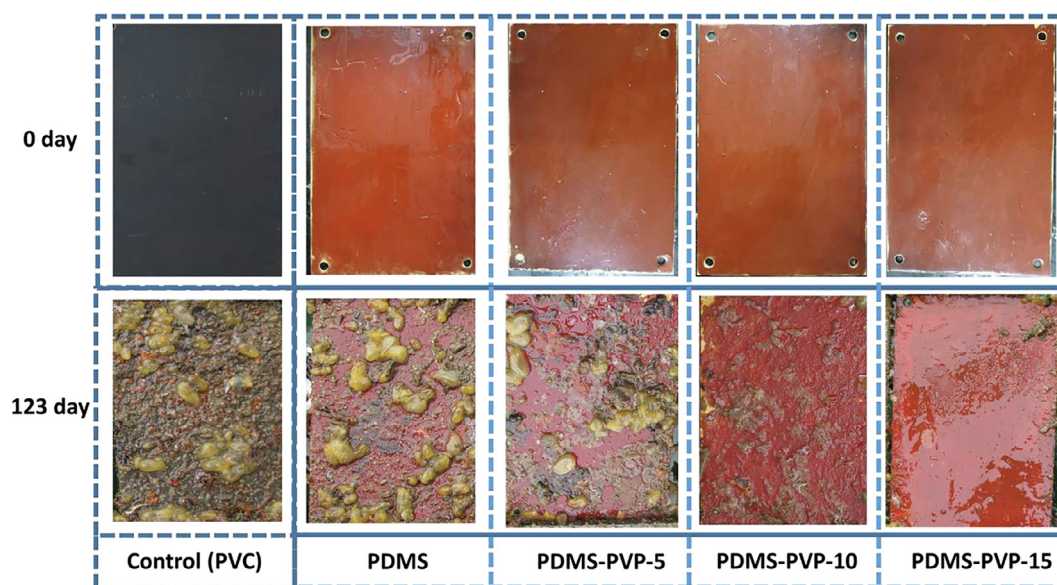


Fig. 10. Images of panels coated by PDMS and PDMS-PVP before and after immersion in seawater for 4 months.

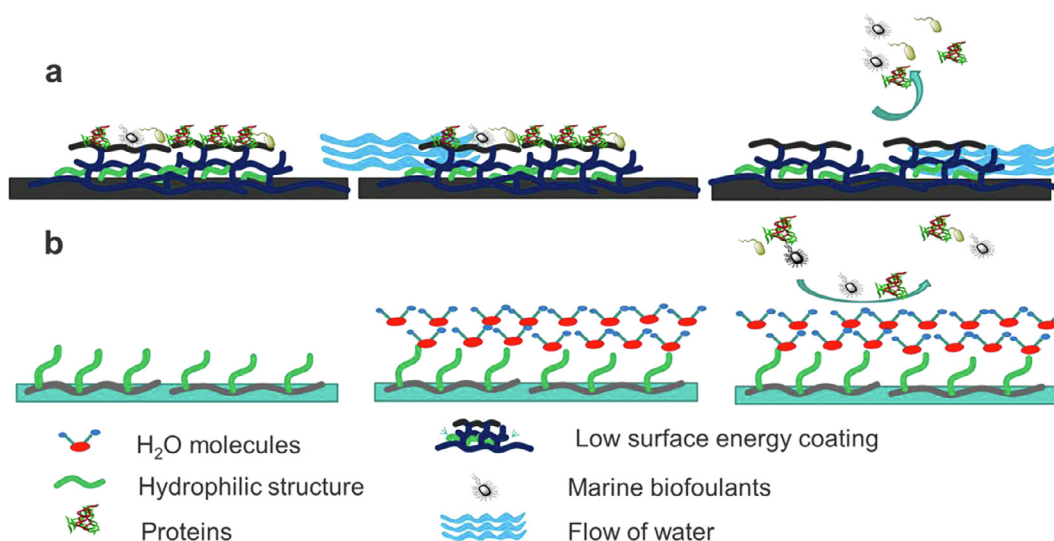


Fig. 11. Schematic illustration of proposed mechanism. (a) PDMS-PVP coating provides the AF/FR surface with low surface energy and (b) hydrophilicity.

extraordinary antifouling capability after a longer time (24 h). This performance was due to that PDMS-PVP coatings exhibited ambiguous surface when exposure to water. These ambiguous surfaces could produce a mosaic of hydrophilic and hydrophobic domains, resulting in the improved FR performance compared to pristine PDMS [61–64].

3.8. Marine field tests

The anti-biofouling performance of PDMS-PVP coating was further evaluated in the marine field (Qingdao, China). Fig. 10 showed the photo images of the panels coated by PDMS and PDMS-PVP-X, which were immersed in seawater for 4 months. A large number of marine organisms (including barnacles, mussels and sponges) could be observed on the surfaces of control PVC panel and PDMS panel. In contrast, the panels coated by PDMS-PVP exhibited significantly improved anti-biofouling performance. Notably, PDMS-PVP-15 sample showed almost no obvious fouling attachment for such a long-term evaluation, indicating this coating was highly stable and promising to be applied in marine anti-biofouling.

3.9. Proposed mechanism

According to these results, a potential mechanism could be proposed:

This novel amphiphilic coating combined the FR properties of PDMS with the hydration layer formed by PVP (Fig. 11). The low surface energy chains of PDMS could induce loose adhesion of biofouling, which was able to be easily washed away. In the marine environment, it could be expected that the resident organisms would be removed from PDMS-PVP coating under dynamic shear stresses during ship moving (fouling-release property) (Fig. 11a). Moreover, in aqueous environment, hydrophilic PVP segments could trap water molecules via preferential interfacial aggregation on the surface, forming a water layer to prevent the biofouling attachment (anti-biofouling property) (Fig. 11b). This mechanism might be the reason of the excellent performance of this novel PDMS-PVP coating even in complicated marine environment.

4. Conclusions

In summary, we developed a novel method to prepare amphiphilic

coatings for the resistance of marine biofouling. These coatings were based on a triethoxysilane-functionalized PVP, which could directly crosslink with PDMS to form an amphiphilic network coating at room temperature. Moreover, the PDMS-PVP coating exhibited excellent anti-biofouling performance in the laboratory and even in marine environment. Because it combined the low surface energy of PDMS and the strong hydrophilicity of PVP. Findings in this work designed an environmentally-friendly anti-biofouling coating, offering a new solution to the problem induced by biofouling in maritime industry.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2019.06.025>.

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